

# Assignment 3: Data Exploration

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## OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

## Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

**TIP:** If your code extends past the page when knit, tidy your code by manually inserting line breaks.

**TIP:** If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

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## Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#import packages
library(tidyverse); library(lubridate); library(here); library(ggplot2)
#check current working directory and check workspace
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#upload ECOTOX dataset, read strings as factors
Neonics <- read.csv(here('./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = TRUE)
#upload Niwot Ridge NEON dataset, read strings as factors
Litter <- read.csv(here('./Data/Raw/NEON_NIW0_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = TRUE)
```

## Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Insects are very important in our ecosystem, related to food webs as well as being important pollinators for human crops, so it's important to understand how these neonicotinoids impact insects, as harmful results could have wider impacts on our ecosystems. This includes how neonicotinoids affect both the target insects that they are meant to treat, and insects that might be harmed unintentionally.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Studying litter and woody debris that falls to the ground in forests likely provides a baseline for understanding forest dynamics (if the forest is healthy, how it changes with varying conditions etc.). Additionally, it can be used to explore carbon and nutrient cycles in forests.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON\\_Litterfall\\_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter is defined as material that is dropped from the forest canopy and has a butt end diameter of <2cm and a length <50cm 2. Litter and fine woody debris samples are collected at terrestrial NEON sites with >2m tall wood vegetation 3. Ground traps are sampled once per year, while elevated traps are sampled on varying time scales based on the type of forest site

## Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#dimensions for Neonics dataset
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#gets summary of effect column in neonics dataset
effect_summ <- summary(Neonics$Effect)

#sort summary to visualize top effects that are studied
sort(effect_summ, decreasing = FALSE)
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)
##           1           5           7           9
##  Biochemistry  Accumulation  Intoxication  Immunological
##          11          12          12          16
##   Morphology      Growth      Enzyme(s)      Genetics
##          22          38          62          82
##   Avoidance      Development  Reproduction Feeding behavior
##         102         136          197         255
##    Behavior      Mortality      Population
##         360         1493         1803
```

Answer: The top three most common effects that are studied are population, mortality, and behavior. These are the most direct ways to see how insects are impacted by neonicotinoids, both population and mortality are easily quantifiable.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#get summary of top 6 most commonly studied species in neonics dataset
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##          667          285          183
##  Carniolan Honey Bee      Bumble Bee      Italian Honeybee
##          152          140          113
##      (Other)
##         3083
```

Answer: The top 6 commonly studied species are all bees, except for the parasitic wasp. Bees are key pollinators, which is crucial for agriculture as well as general global ecosystem health (flowers). Parasitic wasps are also important natural pest control for crops. The health of these species can have great impacts on the performance of agricultural crops, as well as general biodiversity.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
#check class for Neonics 'Conc.1..Author' column
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

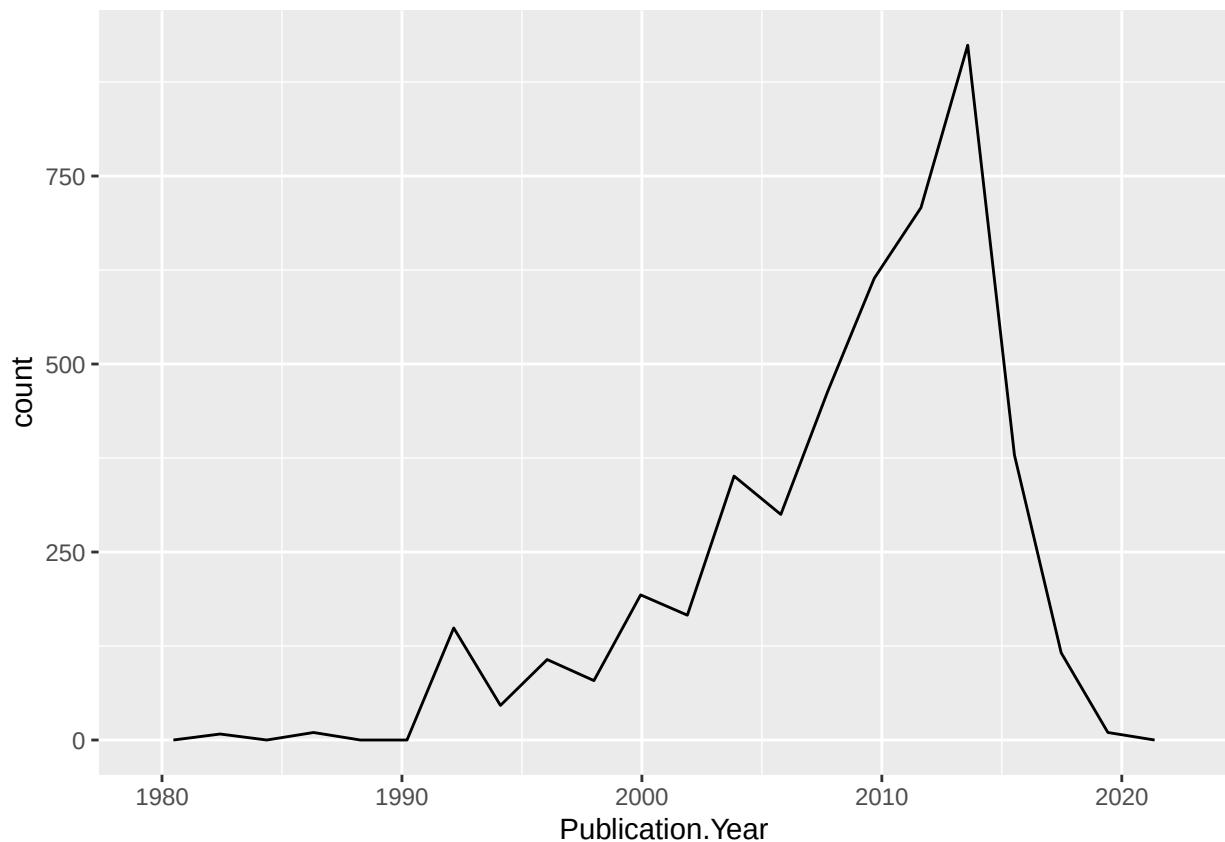
```
#view data frame, commented out for knitting  
#View(Neonics)
```

Answer: It's a factor, likely because there are some entries that are not numeric. For example they include test like, >5.0 or ~0.8.

## Explore your data graphically (Neonics)

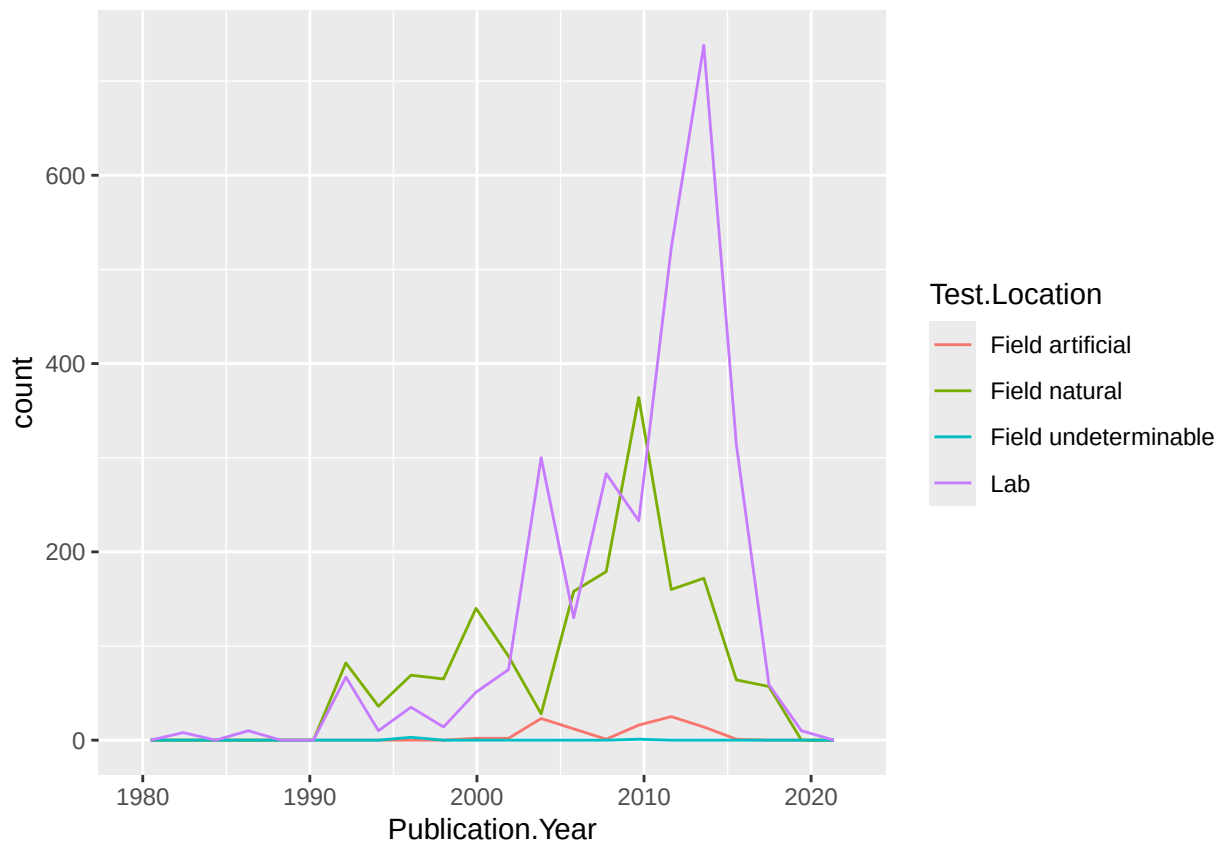
9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
#new plot of Neonics dataset, plotting a frequency polygon graph of the publication year  
ggplot(Neonics) +  
  geom_freqpoly(aes(x = Publication.Year), bins = 20)
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
# create new plot of Neonics dataset, publication year on the x-axis and a separate line for each test  
ggplot(Neonics) +  
  geom_freqpoly(aes(x = Publication.Year, color = Test.Location), bins = 20)
```



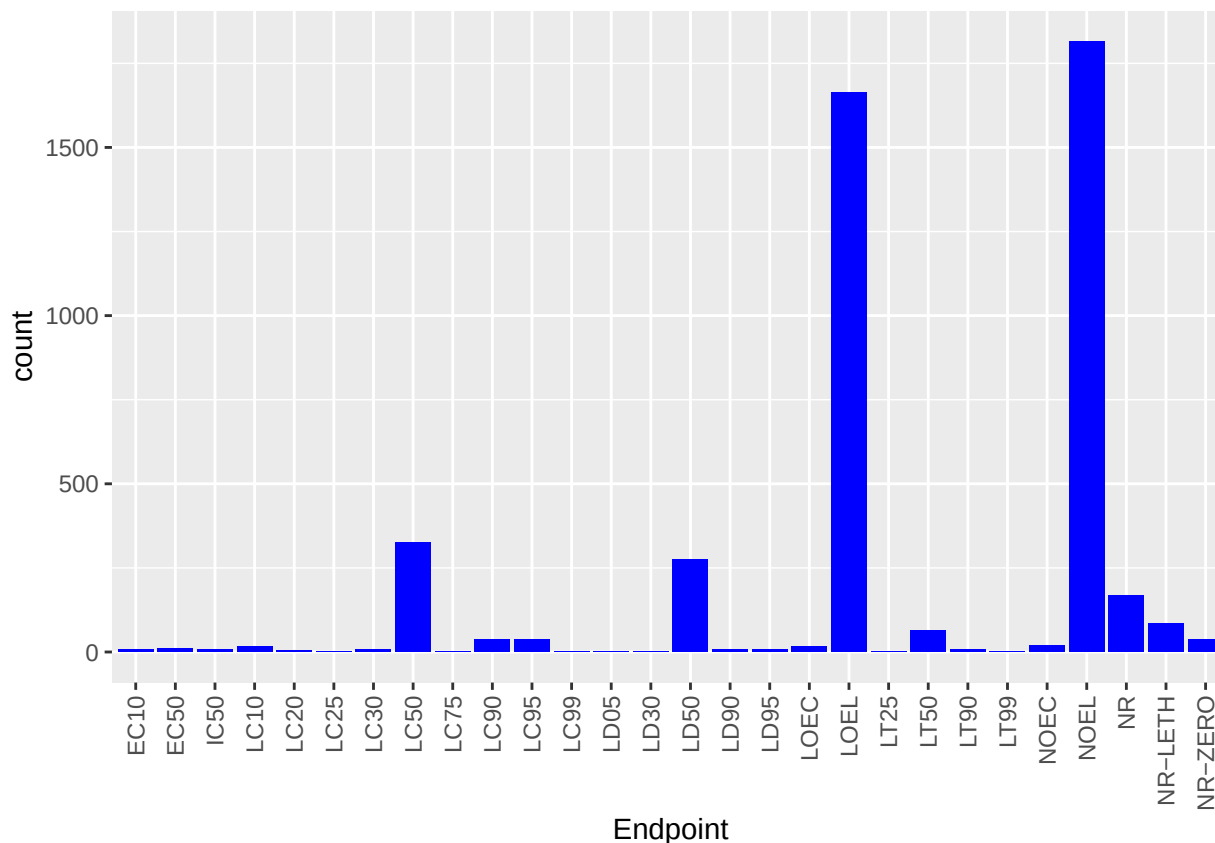
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test location is the Lab. There is variation over time. Lab studies spike around 2003 and again around 2013. Between 1990 and 2000, Field Natural was the most popular, and again in 2009.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX\_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
#create a bar graph of endpoints, rotate x-axis labels using theme()
ggplot(Neonics)+
  geom_bar(aes(x = Endpoint), fill = "blue") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



```
labs(x = "Endpoint")
```

```
## $x
## [1] "Endpoint"
##
## attr("class")
## [1] "labels"
```

Answer: The two most common end points are NOEL and LOEL. NOEL is defined as Terrestrial, no observable effect level, producing effects not significant from responses of controls. LOEL is defined as Terrestrial, lowest observable effect level, lowest dose producing effects that were significant.

## Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#get class of collectDate
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#convert column to date object
Litter$collectDate <- ymd(Litter$collectDate)
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#list the unique values in plotID column
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

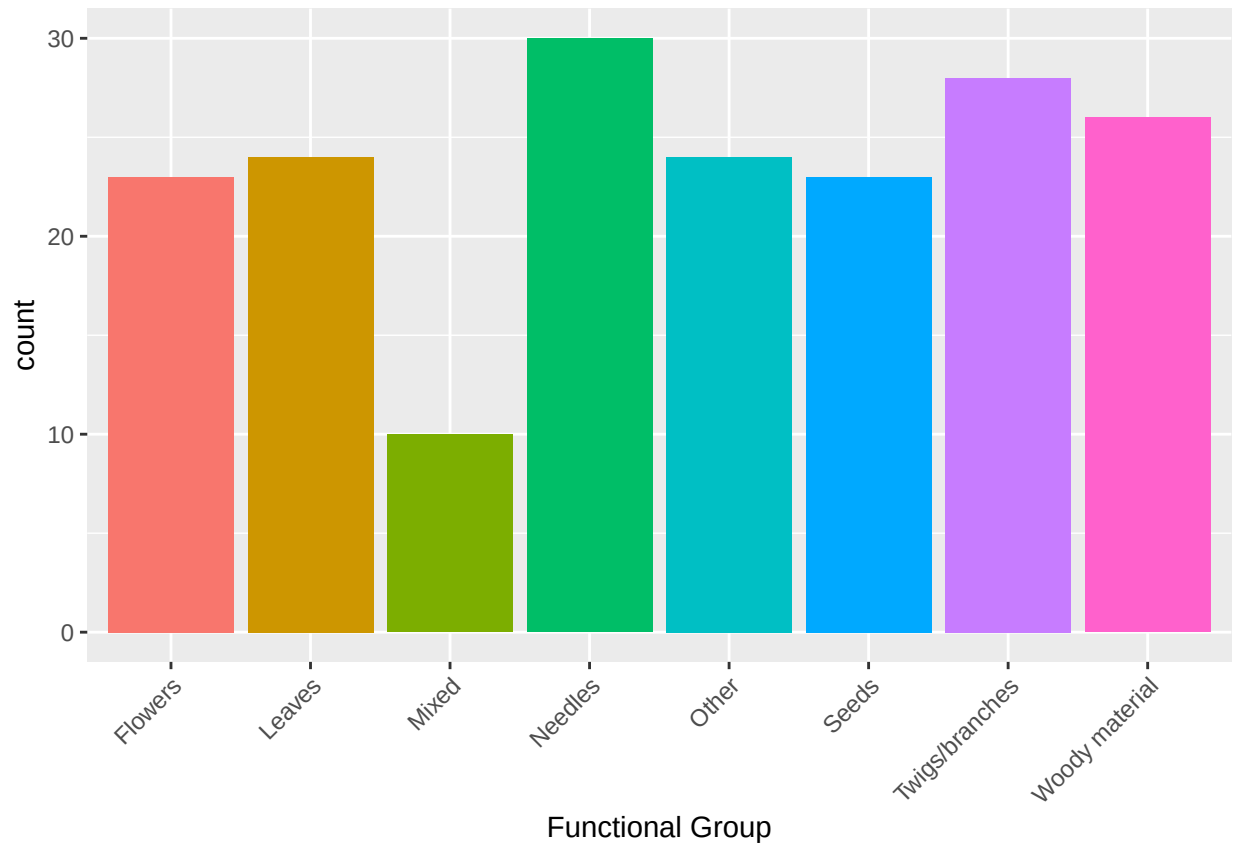
```
#comparison to summary
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14      8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: `Unique` returns the total number of levels used and only the distinct values in the table, while `Summary` gives the count of all of the levels.

14. Create a bar graph of `functionalGroup` counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

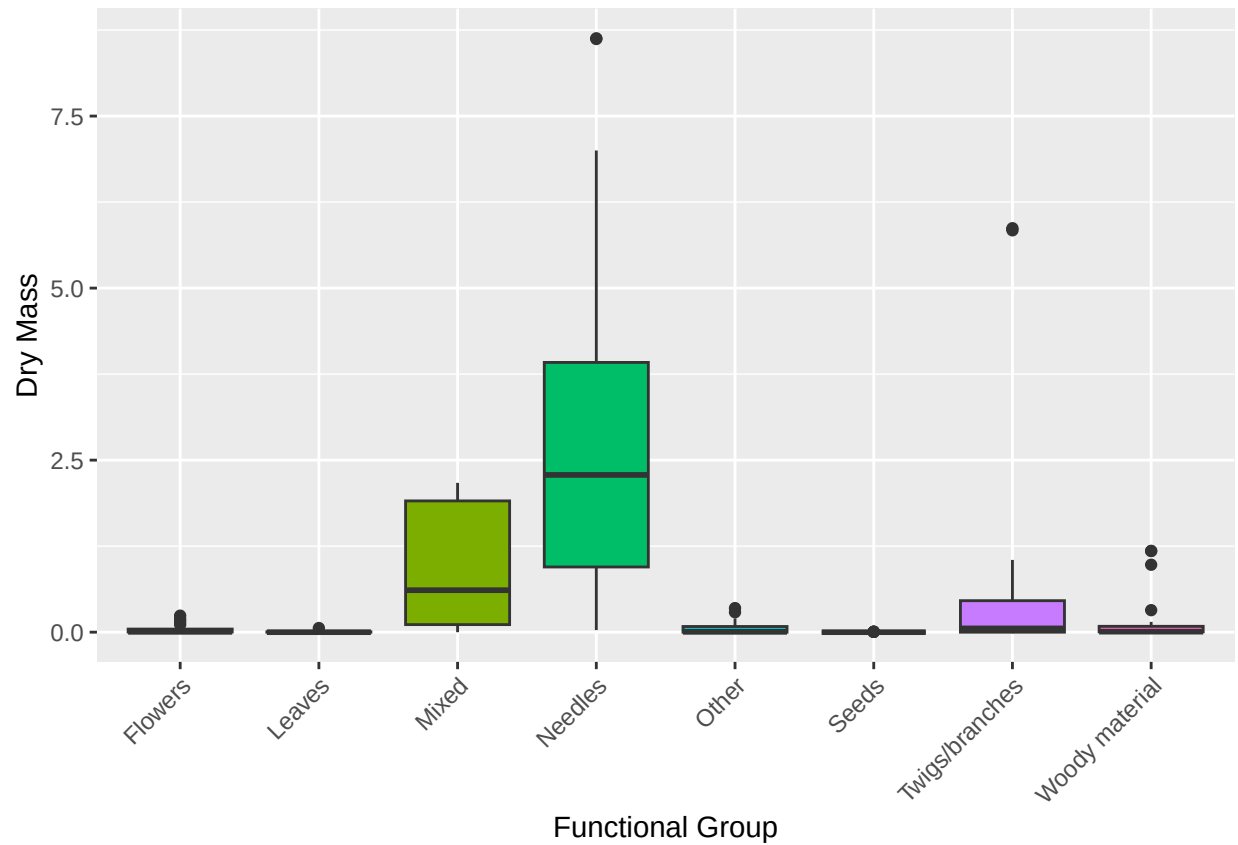
```
#create a bar graph of functionalgroup counts
ggplot(Litter) +
  geom_bar(aes(x = functionalGroup, fill = functionalGroup)) +
  labs(x = "Functional Group") +
  #rotate x-axis text to be more readable
  theme(
    axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  guides(fill = "none") #removes legend
```



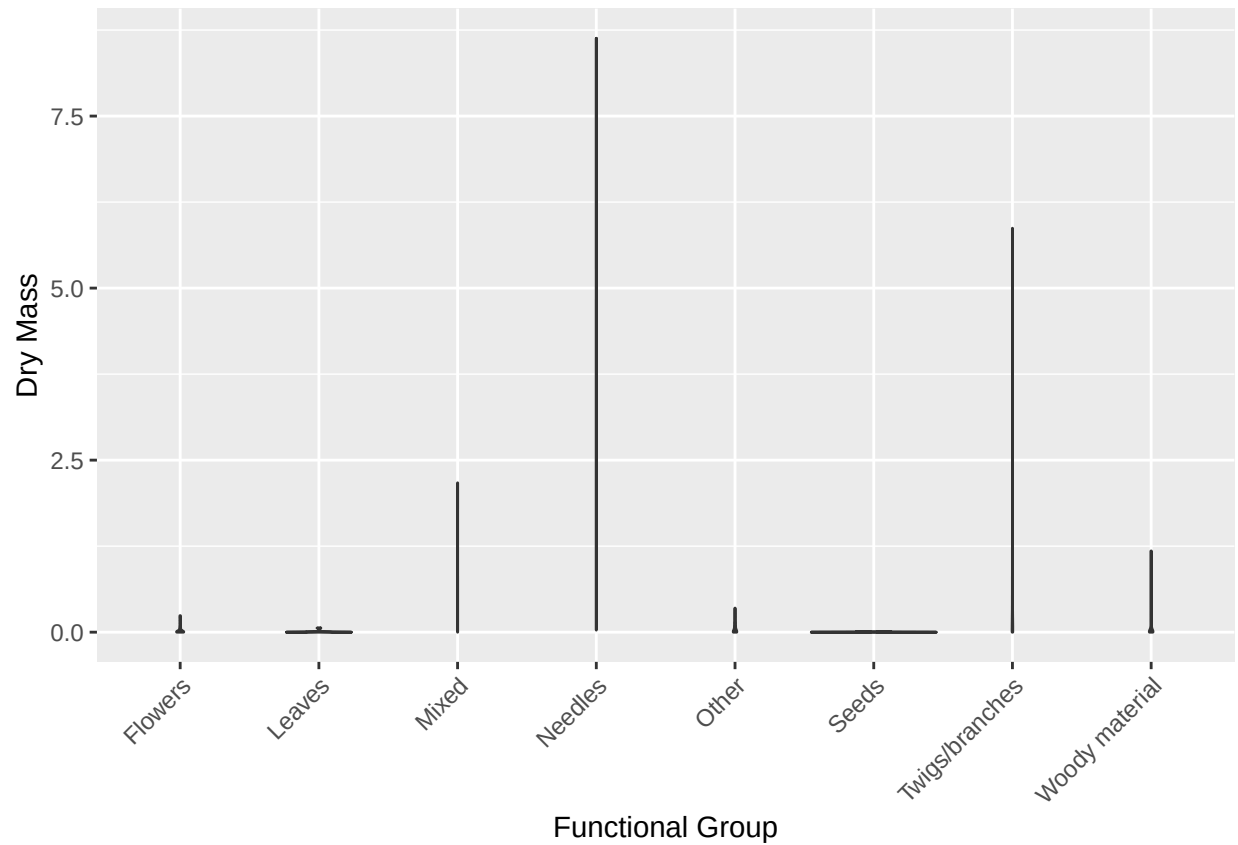
15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by functional-Group.

```
# boxplot of drymass by functional group, testing out putting aes() in ggplot
#makes it easier for copying for violin plot version
ggplot(Litter, aes(x = functionalGroup, y=dryMass, fill= functionalGroup)) +
  geom_boxplot() +
  labs(x = "Functional Group", y = "Dry Mass") +
  #rotate x-axis text to be more readable
  theme(
    axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  guides(fill = "none") #removes legend
```





```
#violin plot of drymass by functional group
ggplot(Litter, aes(x = functionalGroup, y=dryMass, fill= functionalGroup)) +
  geom_violin()+
  labs(x = "Functional Group", y = "Dry Mass") +
#rotate x-axis text to be more readable
theme(
  axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  guides(fill = "none") #removes legend
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The drymass data has a wide range, with some extreme values, which stretches the distribution and makes the violin plot stretched and the shape hard to interpret. The box plot's inclusion of median and quartiles helps better show the spread of values, including extremes/outliers.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles